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Date: ____/____/____

5.4 | Practical application of Friction

Short notes

1) Friction in machines: many machines/instruments have parts that slide against each other, creating frictional forces.

2) Problem with excess friction: • Causes energy loss as work done against the frictional force.
• Converts energy to heat, raising temperature of parts.

3) Benefits of reducing friction: IF frictional forces are minimized, energy loss and temperature rise can be reduced.

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Useful application of friction

1) walking and grip: Friction between shoes and the ground lets you walk without slipping. Thread patterns on tires and soles increase this grip.

2) Brakes in vehicles: Brake pads press against wheels and rotors, and friction slows or stops the vehicle. Without friction, brakes wouldn't work.

3) Writing and holding objects: Friction between pen tip and paper lets you hold objects without them slipping from your hand and write.

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Unwanted effect application of friction.

1) Energy loss in machines: Sliding parts in engines, gears, and bearings waste energy as heat due to friction, lowering efficiency.

2) Wear and tear: Constant friction wears down moving parts like tires, bearings and machine joints, leading to damage and higher maintenance costs.

3) Heat buildup: Excess friction raises temperature in machine parts, which can cause overheating, expansion or failure if not controlled.